1) What is Python? What are the benefits of using Python?

Python is a programming language with objects, modules, threads, exceptions and automatic memory management. The benefits of pythons are that it is simple and easy, portable, extensible, build-in data structure and it is an open source.

2) What is PEP 8?

PEP 8 is a coding convention, a set of recommendation, about how to write your Python code more readable. There is a tool pep8 is a tool to check your Python code against some of the style conventions in PEP 8.

3) What is pickling and unpickling?

Pickle module accepts any Python object and converts it into a string representation and dumps it into a file by using dump function, this process is called pickling. While the process of retrieving original Python objects from the stored string representation is called unpickling.

4) How Python is interpreted?

Python language is an interpreted language. Python program runs directly from the source code. It converts the source code that is written by the programmer into an intermediate language, which is again translated into machine language that has to be executed.

5) What is Python really? You can (and are encouraged) make comparisons to other technologies in your answer

Answer

Here are a few key points: - Python is an interpreted language. That means that, unlike languages like *C* and its variants, Python does not need to be compiled before it is run. Other interpreted languages include *PHP* and *Ruby*.

- Python is dynamically typed, this means that you don't need to state the types of variables when you declare them or anything like that. You can do things like `x=111` and then `x="I'm a string"` without error

- Python is well suited to object orientated programming in that it allows the definition of classes along with composition and inheritance. Python does not have access specifiers (like C++'s `public`, `private`), the justification for this point is given as "we are all adults here"

- in Python, functions are first-class objects. This means that they can be assigned to variables, returned from other functions and passed into functions. Classes are also first class objects

- Writing Python code is quick but running it is often slower than compiled languages. Fortunately, Python allows the inclusion of C based extensions so bottlenecks can be optimised away and often are. The numpy package is a good example of this, it's really quite quick because a lot of the number crunching it does isn't actually done by Python

- Python finds use in many spheres - web applications, automation, scientific modelling, big data applications and many more. It's also often used as "glue" code to get other languages and components to play nice.

- Python makes difficult things easy so programmers can focus on overriding algorithms and structures rather than nitty-gritty low level details.

6) What are the tools that help to find bugs or perform static analysis?

PyChecker is a static analysis tool that detects the bugs in Python source code and warns about the style and complexity of the bug. Pylint is another tool that verifies whether the module meets the coding standard.

7) What is the output of print str[2:5] if str = 'Hello World!'?

It will print characters starting from 3rd to 5th. Output would be llo.

8) What is the difference between list and tuple?

The difference between list and tuple is that list is mutable while tuple is not. Tuple can be hashed for e.g as a key for dictionaries.

9) How are arguments passed by value or by reference?

Arguments are passed by assignment. The rationale behind this is twofold:

1. the parameter passed in is actually a *reference* to an object (but the reference is passed by value)

2. some data types are mutable, but others aren't

So:

- If you pass a *mutable* object into a method, the method gets a reference to that same object and you can mutate it to your heart's delight, but if you rebind the reference in the method, the outer scope will know nothing about it, and after you're done, the outer reference will still point at the original object.

- If you pass an *immutable* object to a method, you still can't rebind the outer reference, and you can't even mutate the object.

10) What is the output of print str + "TEST" if str = 'Hello World!'?

It will print concatenated string. Output would be Hello World!TEST.

11) What are the built-in type does python provides?

There are mutable and Immutable types of Pythons built in types Mutable built-in types

- List
- Sets
- Dictionaries

Immutable built-in types

- Strings
- Tuples
- Numbers

12) What is namespace in Python?

Generally speaking, a **namespace** (sometimes also called a context) is a naming system for making names unique to avoid ambiguity. An identifier defined in a namespace is associated with that namespace. This way, the same identifier can be independently defined in multiple namespaces. (Like the same file names in different directories)

13) What is lambda in Python?

It is a single expression anonymous function often used as inline function.

14) What is the output of print list[2:] if list = ['abcd', 786 , 2.23, 'john', 70.2]?

It will print elements starting from 3rd element. Output would be [2.23, 'john', 70.200000000000003].

15) What is pass in Python?

Pass means, no-operation Python statement, or in other words it is a place holder in compound statement, where there should be a blank left and nothing has to be written there.

16) In Python what are iterators?

In Python, iterators are used to iterate a group of elements, containers like list.

17) What is unittest in Python?

A unit testing framework in Python is known as unittest. It supports sharing of setups, automation testing, shutdown code for tests, aggregation of tests into collections etc.

18) In Python what is slicing?

A mechanism to select a range of items from sequence types like list, tuple, strings etc. is known as slicing.

19) What is the output of print tinylist * 2 if tinylist = [123, 'john']?

It will print list two times. Output would be [123, 'john', 123, 'john'].

20) What is docstring in Python?

A Python documentation string is known as docstring, it is a way of documenting Python functions, modules and classes. It is a string literal that occurs as the first statement in a module, function, class, or method definition. Such a docstring becomes the __doc__ special attribute of that object.

21) How can you copy an object in Python?

To copy an object in Python, you can try copy.copy( ) or copy.deepcopy( ) for the general case. You cannot copy all objects but most of them.

22) What is negative index in Python?

Python sequences can be indexed in positive and negative numbers. For positive index, 0 is the first index, 1 is the second index and so forth. For negative index, (-1) is the last index and (-2) is the second last index and so forth.

23) How you can convert a number to a string?

In order to convert a number into a string, use the inbuilt function str( ). If you want a octal or hexadecimal representation, use the inbuilt function oct( ) or hex( ).

24) What is the difference between xrange( ) and range( )?

range( ) creates a list, so if you do range(1, 10000000) it creates a list in memory with 9999999 elements. xrange( ) is a sequence object that evaluates lazily. So xrange( ) is faster

25) What is module and package in Python?

In Python, module is the way to structure program. Each Python program file is a module, which imports other modules like objects and attributes.

The folder of Python program is a package of modules. A package can have modules or subfolders.

26) Mention what are the rules for local and global variables in Python?

Local variables: If a variable is assigned a new value anywhere within the function's body, it's assumed to be local.

Global variables: Those variables that are only referenced inside a function are implicitly global.

27) How can you share global variables across modules?

To share global variables across modules within a single program, create a special module. Import the config module in all modules of your application. The module will be available as a global variable across modules.

28) Explain how can you make a Python Script executable on Unix?

To make a Python Script executable on Unix, you need to do two things,

- Script file's mode must be executable and
- add #!/usr/bin/python as first line of your code

29) Explain how to delete a file in Python?

By using a command os.remove (filename) or os.unlink(filename)

30) Explain how can you generate random numbers in Python?

To generate random numbers in Python, you need to import module random as

import random

random.random()

This returns a random floating point number in the range [0,1)

31) What is the difference between tuples and lists in Python?

The main differences between lists and tuples are – Lists are enclosed in brackets ([ ]) and their elements and size can be changed, while tuples are enclosed in parentheses (( )) and cannot be updated. Tuples can be thought of as read-only lists.

32) Mention the use of // operator in Python?

It is a Floor Divisionoperator , which is used for dividing two operands with the result as quotient showing only digits before the decimal point. For instance, 10//5 = 2 and 10.0//5.0 = 2.0.

33) Mention five benefits of using Python?

- Python comprises of a huge standard library for most Internet platforms like Email, HTML, etc.

- Python does not require explicit memory management as the interpreter itself allocates the memory to new variables and free them automatically
- Provide easy readability due to use of square brackets and indentation
- Easy-to-learn for beginners
- Having the built-in data types saves programming time and effort from declaring variables

34) Mention the use of the split function in Python?

The use of the split function in Python is that it breaks a string into shorter strings using the defined separator. It gives a list of all words present in the string.

35) What are Python's dictionaries?

Python's dictionaries are kind of hash table type. They work like associative arrays or hashes found in Perl and consist of key-value pairs. A dictionary key can be almost any Python type, but are usually numbers or strings. Values, on the other hand, can be any arbitrary Python object. Dictionaries are enclosed by curly braces ({ }) and values can be assigned and accessed using square braces ([]). Ex-

```
dict = {}
dict['one'] = "This is one"
dict[2]     = "This is two"
tinydict = {'name': 'john','code':6734, 'dept': 'sales'}
```

36) How will you get all the keys and values from the dictionary?

Usinchr(x) − Converts an integer to a character.g dictionary.keys() function, we can get all the keys from the dictionary object.

```
print dict.keys()   # Prints all the keys
```

Using dictionary.values() function, we can get all the values from the dictionary object.

```
print dict.values()   # Prints all the values
```

37) How will you convert an integer to a character in python?
chr(x) − Converts an integer to a character.

38) How will you convert a single character to its integer value (ASCII) in python?
ord(x) − Converts a single character to its integer value.

39) How will you get the length of the string?

len(string) – Returns the length of the string.

40) What is the output of -- 3 in [1, 2, 3]?
True

41) How will you compare two lists?
cmp(list1, list2) – Compares elements of both lists.

42) How will you reverse a list?
list.reverse() – Reverses objects of list in place.